Parallel and HPC
HW7 Report

Link to GITHUB repository:

https://github.com/DMelkonyan111/Parallel-and-HPC-Homeworks

Fibonacci solution

For this assignment, the program executes in the following way. The main function creates the variable num, which holds the Fibonacci number (index, but starting from 1), calls the fib function and prints the output. In the fib function there is the base case check if num is less than or equal to 2, in which case 1 is returned, then there is the sequential algorithm if num is not more than 10, which is better than the parallel version as num is small, and then there is the parallelized recursion. In the omp parallel region with THREAD_NUM number of threads in an omp single block the recursive function fib_recursion is called which makes 2 variables x and y and their values are the returned values from the recursion with (num – 1) and (num – 2) arguments, and the implementations of recursion are set up with tasks. There is a taskwait after x and y are found to avoid data race and eventually x + y is returned.

```
davitm@ubuntu-aua-os:~/workshop/Homeworks/7$ gcc -fopenmp fibonacci.c -o fib
davitm@ubuntu-aua-os:~/workshop/Homeworks/7$ ./fib
The fibonacci number at 15's position is 610
davitm@ubuntu-aua-os:~/workshop/Homeworks/7$ gcc -fopenmp fibonacci.c -o fib
davitm@ubuntu-aua-os:~/workshop/Homeworks/7$ ./fib
The fibonacci number at 20's position is 6765
davitm@ubuntu-aua-os:~/workshop/Homeworks/7$ gcc -fopenmp fibonacci.c -o fib
davitm@ubuntu-aua-os:~/workshop/Homeworks/7$ ./fib
The fibonacci number at 25's position is 75025
davitm@ubuntu-aua-os:~/workshop/Homeworks/7$
```